

Lab 3 Architecture Justification Coffee Kiosk System

Selected architecture: Three-Layer Architecture

1. Architectural Choice

The selected style is Three-Layer Architecture, organised into a Presentation Layer (User Interface Component), Business Layer (Order Manager, Payment Service, and Receipt Printer Components), and Data Layer (Database Component). This provides clear separation of concerns and defines a structured flow between user interaction, business logic, and stored data.

2. Reasons for Selection

Natural fit for the kiosk workflow. The kiosk follows a straightforward sequence: the customer selects a drink and size through the touch screen; the Order Manager validates the order and coordinates payment and receipt printing; and the Database provides menu and pricing information. This simple workflow maps naturally to a layered structure without the additional deployment and communication complexity of Microservices or Client-Server architecture.

Low-complexity, self-contained system. The kiosk has a fixed menu of six options (3 coffee types × 2 sizes) and serves a relatively simple ordering process. It does not require independent scaling of individual services or a separate centralized server. Keeping the components within a single kiosk system reduces operational complexity and communication overhead.

3. Security Advantage

The Presentation Layer does not directly access the Data Layer. Requests from the touch screen pass through the Order Manager in the Business Layer, which can validate item selections, sizes, and payment information before accessing the database. This controlled access path helps prevent malformed or unauthorized requests from directly reaching the database.

4. Performance Benefit

The components can communicate locally without requiring network round-trips between layers. This reduces communication overhead and helps maintain responsive touch-screen interactions.